

# Collision Ideals and Off-Diagonal Sheets

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## Abstract

For a polynomial map  $F = (F_1, \dots, F_n): \mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^n$ , define in  $S = \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n]$  the collision and diagonal ideals

$$I_R = (F_i(x) - F_i(y))_{i=1}^n, \quad I_{\Delta} = (x_i - y_i)_{i=1}^n.$$

The inclusion  $I_R \subseteq I_{\Delta}$  defines the obstruction module  $I_{\Delta}/I_R$ , and

$$I_{\Delta}/I_R = 0 \iff I_R = I_{\Delta} \iff F \text{ is a polynomial automorphism.}$$

In dimension two, let  $\mathcal{D}_F$  be the finite-cover diagram of a map with constant nonzero Jacobian determinant, and let  $\mathcal{H}(\mathcal{D}_F)$  be its hidden-inertia locus. Then

$$\mathcal{H}(\mathcal{D}_F) = \emptyset \implies I_R = I_{\Delta}.$$

For the separable nonnormal cubic class in dimension three, let

$$K = \mathbb{C}(F_1, F_2, F_3), \quad L = \mathbb{C}(x_1, x_2, x_3),$$

and let  $N$  be the normal closure of  $L/K$ . Then

$$L \otimes_K L \simeq L \times N, \quad \text{Gal}(N/K) \simeq S_3.$$

The off-diagonal factor  $N$  is the ordered-root Galois-closure sheet. Consequently,  $I_{\Delta}/I_R \neq 0$ . These constructions and implications are formalized in Lean 4, with the external inputs isolated explicitly.

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# 1 General construction

## 1.1 The two ideals and their obstruction

Throughout the paper, the base field is  $\mathbb{C}$ . Let

$$F = (F_1, \dots, F_n): X = \mathbb{A}_{\mathbb{C}}^n \longrightarrow \mathbb{A}_{\mathbb{C}}^n$$

be a polynomial map, and let

$$S = \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n].$$

The map  $F$  is Keller if

$$\det JF \in \mathbb{C}^\times.$$

Define

$$I_R(F) = (F_i(x) - F_i(y))_{i=1}^n, \quad I_\Delta = (x_i - y_i)_{i=1}^n, \quad \text{Obs}(F) := I_\Delta / I_R(F).$$

For a fixed map  $F$ , we abbreviate  $I_R(F)$  to  $I_R$ .

**Proposition 1.1** (Canonical containment). *For every polynomial map  $F$ ,*

$$I_R(F) \subseteq I_\Delta.$$

*Proof.* For  $H \in \mathbb{C}[x_1, \dots, x_n]$ , the difference  $H(x) - H(y)$  belongs to  $(x_1 - y_1, \dots, x_n - y_n)$ . This follows by induction on  $H$ , or by telescoping each monomial one coordinate at a time. Apply the statement to each coordinate  $F_i$ .  $\square$

Let  $C_F := S/I_R$  and  $A := \mathbb{C}[x_1, \dots, x_n]$ . Diagonal evaluation induces a surjective ring map

$$\bar{\mu}_F: C_F \longrightarrow A, \quad [s]_{I_R} \longmapsto s(x, x).$$

**Proposition 1.2** (The diagonal kernel). *As ideals of  $C_F$ ,*

$$\ker(\bar{\mu}_F) = \text{Obs}(F) = I_\Delta / I_R.$$

Consequently,

$$\ker(\bar{\mu}_F) = 0 \iff \text{Obs}(F) = 0 \iff I_R = I_\Delta.$$

*Proof.* The kernel of diagonal evaluation  $S \rightarrow A$  is  $I_\Delta$ . Passing through  $S/I_R$  therefore identifies the displayed kernel with  $I_\Delta/I_R$ . Since  $I_R \subseteq I_\Delta$ ,

$$I_\Delta/I_R = 0 \iff I_R = I_\Delta.$$

□

## 1.2 The fiber-product interpretation

Let

$$A = \mathbb{C}[x_1, \dots, x_n], \quad B = \mathbb{C}[F_1, \dots, F_n] \subseteq A.$$

Let  $X = \operatorname{Spec} A$ , let  $Y = \operatorname{Spec} B$ , and let

$$f_F: X \longrightarrow Y$$

be the morphism induced by  $B \hookrightarrow A$ . The homomorphisms  $\iota_x, \iota_y: A \rightrightarrows C_F$ , defined by  $\iota_x(a) = [a(x)]$  and  $\iota_y(a) = [a(y)]$ , agree on  $B$  because  $[F_i(x)] = [F_i(y)]$  in  $C_F$ . Their common restriction equips  $C_F$  with a  $B$ -algebra structure.

**Theorem 1.3** (Fiber-product interpretation). *There is a canonical  $B$ -algebra equivalence and its dual affine-scheme equivalence*

$$C_F \cong A \otimes_B A, \quad R_F := \operatorname{Spec} C_F \cong X \times_Y X.$$

*Under the first equivalence,  $\bar{\mu}_F$  is tensor multiplication*

$$A \otimes_B A \longrightarrow A, \quad a \otimes a' \longmapsto aa'.$$

*Its dual morphism is the diagonal*

$$\Delta_{X/Y}: X \longrightarrow X \times_Y X.$$

*Proof.* A map  $C_F \rightarrow T$  to a commutative  $\mathbb{C}$ -algebra  $T$  is equivalent to a pair of maps  $f, g: A \rightrightarrows T$  satisfying

$$f(F_i) = g(F_i) \quad (1 \leq i \leq n).$$

Since the  $F_i$  generate  $B$ , this is precisely the universal property of  $A \otimes_B A$ . Applying  $\operatorname{Spec}$  to this equivalence and using

$$\operatorname{Spec}(A \otimes_B A) \cong \operatorname{Spec} A \times_{\operatorname{Spec} B} \operatorname{Spec} A$$

gives the asserted fiber-product equivalence. The statements about multiplication and the diagonal follow on pure tensors. □

$$R_F(\mathbb{C}) = \{(u, v) \in X(\mathbb{C}) \times X(\mathbb{C}) : F(u) = F(v)\}.$$

Inside

$$X \times X = \operatorname{Spec} S,$$

the ideals  $I_R$  and  $I_\Delta$  define, respectively,  $X \times_Y X$  and the image of  $\Delta_{X/Y}$ . The contravariance of  $I \mapsto V(I)$  therefore gives the equivalences

$$I_R \subseteq I_\Delta \iff \Delta_{X/Y}(X) \subseteq X \times_Y X$$

and

$$I_R = I_\Delta \iff X \times_Y X = \Delta_{X/Y}(X)$$

as closed subschemes of  $X \times X$ . Thus the algebraic and geometric gaps are two descriptions of the same scheme-theoretic containment.

### 1.3 The off-diagonal factor and secant projectors

For every polynomial map  $F$ , define

$$I_{\text{off}} := I_R : I_\Delta, \quad C_F^\circ := S/I_{\text{off}}, \quad R_F^\circ := \text{Spec } C_F^\circ.$$

This definition does not require the diagonal to be clopen.

**Corollary 1.4** (Off-diagonal emptiness criterion). *For every polynomial map  $F$ , the following are equivalent:*

$$R_F^\circ = \emptyset \iff C_F^\circ = 0 \iff \text{Obs}(F) = 0 \iff I_R = I_\Delta.$$

*Proof.*

$$R_F^\circ = \emptyset \iff C_F^\circ = 0 \iff I_R : I_\Delta = S.$$

Moreover,

$$I_R : I_\Delta = S \iff I_\Delta \subseteq I_R \iff I_R = I_\Delta.$$

The equivalence with  $\text{Obs}(F) = 0$  is Proposition 1.2. □

The idempotent mechanism behind a clopen diagonal is purely commutative-algebraic. Let  $C$  be a commutative ring,  $J \subseteq C$  an ideal,  $\delta \in C$ , and  $c \in C^\times$ . Suppose

$$\delta J = 0, \quad \delta - c \in J.$$

Define

$$q := 1 - c^{-1}\delta.$$

**Theorem 1.5** (Abstract off-diagonal projector). *The element  $q$  is idempotent and*

$$J = Cq, \quad \text{Ann}_C(J) = C(1 - q).$$

*In particular,*

$$J = 0 \iff q = 0.$$

*Proof.* The congruence  $\delta \equiv c \pmod{J}$  gives  $q \in J$ . For every  $j \in J$ ,

$$qj = j - c^{-1}\delta j = j.$$

Taking  $j = q$  proves  $q^2 = q$ ; the same identity gives  $J = Cq$ . Finally,

$$aq = 0 \iff a = a(1 - q),$$

so  $\text{Ann}_C(J) = \text{Ann}_C(Cq) = C(1 - q)$ . □

Whenever the theorem is applied with

$$C = C_F, \quad J = \text{Obs}(F),$$

the identities  $J = C_F q$  and  $C_F/J \cong A$  give

$$C_F \cong A \times C_F/C_F(1 - q).$$

Thus the first factor is the diagonal and the second is its off-diagonal complement.

## 1.4 Generic fibers and marked-root decomposition

Let  $F$  be dominant and generically finite, and let

$$A = \mathbb{C}[x_1, \dots, x_n], \quad B = \mathbb{C}[F_1, \dots, F_n] \subseteq A.$$

Let

$$K = \text{Frac}(B), \quad L = \text{Frac}(A).$$

Let

$$\theta: K \otimes_B A \longrightarrow L, \quad \lambda \otimes a \longmapsto \lambda a.$$

**Theorem 1.6** (Generic collision bridge). *The map  $\theta$  is an isomorphism. Consequently, there is a  $K$ -algebra isomorphism*

$$K \otimes_B C_F \cong L \otimes_K L,$$

and the following square commutes:

$$\begin{array}{ccc} K \otimes_B C_F & \xrightarrow{1 \otimes \bar{\mu}_F} & K \otimes_B A \\ \sim \downarrow & & \downarrow \theta \sim \\ L \otimes_K L & \xrightarrow{\mu_L} & L. \end{array}$$

*Proof.* Let  $U = B \setminus \{0\}$ . Since  $K = U^{-1}B$ , localization gives

$$K \otimes_B A \xrightarrow{\sim} U^{-1}A, \quad \frac{b}{u} \otimes a \longmapsto \frac{ba}{u}.$$

Because  $B \subseteq A$  and  $A$  is a domain,

$$U = B \setminus \{0\} \subseteq A \setminus \{0\}, \quad U^{-1}A \hookrightarrow \text{Frac}(A) = L, \quad \frac{a}{u} \longmapsto \frac{a}{u}.$$

Composing this injection with the displayed isomorphism gives  $\theta$ , so  $\theta$  is injective.

Since  $F$  is generically finite,  $L/K$  is a finite extension. The image of  $U^{-1}A$  in  $L$  is

$$K[x_1, \dots, x_n].$$

Each  $x_i \in L$  is algebraic over  $K$ , so this is a finite-dimensional  $K$ -algebra. As a finite-dimensional  $K$ -domain, it is a field: multiplication by a nonzero element is an injective endomorphism of a finite-dimensional  $K$ -vector space and therefore is surjective. Because it contains  $A$ , it contains  $\text{Frac}(A) = L$ ; the reverse inclusion is already given by the embedding above. Therefore

$$U^{-1}A = L,$$

and  $\theta$  is an isomorphism.

By Theorem 1.3, identify  $C_F$  with  $A \otimes_B A$ . There is a  $K$ -algebra isomorphism

$$\beta: K \otimes_B (A \otimes_B A) \xrightarrow{\sim} (K \otimes_B A) \otimes_K (K \otimes_B A)$$

defined on pure tensors by

$$\beta(\lambda \otimes (a \otimes a')) = (\lambda \otimes a) \otimes (1 \otimes a').$$

Its inverse is

$$(\lambda \otimes a) \otimes (\lambda' \otimes a') \mapsto \lambda \lambda' \otimes (a \otimes a').$$

The balancing relations over  $B$  and  $K$  show that these formulas are well-defined, and the two displayed maps are mutually inverse. Composing  $\beta$  with  $\theta \otimes_K \theta$  gives

$$\Phi_F: K \otimes_B C_F \xrightarrow{\sim} L \otimes_K L.$$

It remains to identify the diagonal map. Under  $C_F \cong A \otimes_B A$ , the homomorphism  $\bar{\mu}_F$  is multiplication. Therefore, on a pure tensor,

$$\begin{aligned} (\mu_L \circ \Phi_F)(\lambda \otimes (a \otimes a')) &= (\lambda a) a' = \lambda a a', \\ (\theta \circ (1 \otimes \bar{\mu}_F))(\lambda \otimes (a \otimes a')) &= \theta(\lambda \otimes a a') = \lambda a a'. \end{aligned}$$

Pure tensors generate  $K \otimes_B C_F$ , so the two homomorphisms are equal. This proves the commutative square.  $\square$

Suppose now that  $L/K$  is separable and has a power basis  $L = K(\alpha)$ . Let  $m_\alpha(T) \in K[T]$  be the minimal polynomial of  $\alpha$ , and let  $h_\alpha(T) \in L[T]$  be determined by

$$m_\alpha(T) = (T - \alpha)h_\alpha(T) \quad \text{in } L[T].$$

**Theorem 1.7** (Marked-root tensor decomposition). *Give  $L \otimes_K L$  its  $L$ -algebra structure through the left tensor factor. There is an  $L$ -algebra equivalence*

$$L \otimes_K L \cong L \times L[T]/(h_\alpha),$$

*under which tensor multiplication  $L \otimes_K L \rightarrow L$  is projection onto the first factor.*

*Proof.* The power-basis presentation

$$L \cong K[T]/(m_\alpha)$$

and base change from  $K$  to  $L$  give an  $L$ -algebra isomorphism

$$L \otimes_K L \xrightarrow{\sim} L[T]/(m_\alpha), \quad 1 \otimes \alpha \mapsto T.$$

Since  $m_\alpha$  is separable,  $m'_\alpha(\alpha) \neq 0$ . Differentiating

$$m_\alpha(T) = (T - \alpha)h_\alpha(T)$$

and evaluating at  $T = \alpha$  gives

$$h_\alpha(\alpha) = m'_\alpha(\alpha) \neq 0.$$

Thus  $T - \alpha$  does not divide  $h_\alpha$ , so these two polynomials are coprime in  $L[T]$ . The Chinese remainder theorem now gives

$$L[T]/(m_\alpha) \xrightarrow{\sim} L[T]/(T - \alpha) \times L[T]/(h_\alpha) \xrightarrow{\sim} L \times L[T]/(h_\alpha).$$

Under the first isomorphism above, a pure tensor  $\ell \otimes p(\alpha)$ , with  $p \in K[T]$ , corresponds to the class of  $\ell p(T)$ . Tensor multiplication sends it to  $\ell p(\alpha)$ , which is evaluation at  $T = \alpha$ . Under the Chinese remainder isomorphism, this evaluation map is projection onto the factor  $L[T]/(T - \alpha) \cong L$ . Hence tensor multiplication is the first projection.  $\square$

Together with Theorem 1.6, this gives

$$K \otimes_B C_F \cong L \times L[T]/(h_\alpha),$$

where the first projection is the generic diagonal.

## 1.5 Normalized covers and inertia

Let  $L/K$  be finite and separable, and let  $N/K$  be its normal closure, with a fixed embedding  $L \hookrightarrow N$ . Let

$$G = \text{Gal}(N/K), \quad H = \text{Gal}(N/L).$$

**Proposition 1.8** (Conjugate-embedding action). *Restriction induces a bijection*

$$G/H \xrightarrow{\sim} \text{Hom}_K(L, N), \quad gH \mapsto g|_L.$$

*The induced action of  $G$  on  $G/H$  is faithful; equivalently,*

$$\text{core}_G(H) = \bigcap_{g \in G} gHg^{-1} = 1.$$

*Proof.* For  $g_1, g_2 \in G$ ,

$$g_1|_L = g_2|_L \iff g_2^{-1}g_1 \in H \iff g_1H = g_2H.$$

Every  $K$ -embedding  $L \hookrightarrow N$  extends to an element of  $G$ , so restriction gives the stated bijection. Its action kernel is

$$\ker(G \curvearrowright G/H) = \bigcap_{g \in G} gHg^{-1} = \text{core}_G(H).$$

If  $\sigma \in \text{core}_G(H)$ , then

$$\sigma|_{g(L)} = \text{id}_{g(L)} \quad (g \in G).$$

Since  $N$  is the normal closure,

$$N = K(g(L) : g \in G),$$

and therefore  $\sigma = \text{id}_N$ .  $\square$

Let

$$Y = \text{Spec } B, \quad \overline{X} = \text{Norm}_L(Y), \quad Z = \text{Norm}_N(Y).$$

The marked embedding  $L \hookrightarrow N$  induces a canonical commutative triangle

$$Z \xrightarrow{\pi} \overline{X} \xrightarrow{\bar{\nu}} Y, \quad \nu = \bar{\nu} \circ \pi : Z \longrightarrow Y.$$

Suppose that  $\bar{\nu}$  and  $\nu$  are finite, and fix an open immersion over  $Y$

$$j: X = \operatorname{Spec} A \hookrightarrow \bar{X}$$

with  $\bar{\nu} \circ j = f_F$ . Denote this diagram and its deleted boundary by

$$\mathcal{D} = \left( Z \xrightarrow{\pi} \bar{X} \xrightarrow{\bar{\nu}} Y, j: X \hookrightarrow \bar{X} \right), \quad \partial X = \bar{X} \setminus j(X).$$

Fix  $E \in Z^{(1)}$ . Let  $w_E$  be its divisorial valuation, let  $b = \nu(E)$ , and let  $v_b = w_E|_K$ . Let  $D_E \leq G$  and  $I_E \trianglelefteq D_E$  be its decomposition and inertia groups. Let

$$\mathcal{Q}_E := D_E \backslash G/H.$$

For  $\omega = D_E g H \in \mathcal{Q}_E$ , let

$$u_\omega = (g^{-1} w_E)|_L, \quad \operatorname{cent}(E, \omega) = \operatorname{cent}_{\bar{X}}(u_\omega),$$

and let

$$i_E(\omega) = [I_E : I_E \cap g H g^{-1}].$$

**Proposition 1.9** (Conjugate divisors and double cosets). *The assignments above are well-defined, and restriction of valuations induces a bijection*

$$\mathcal{Q}_E \xrightarrow{\sim} \{\text{prime divisors of } \bar{X} \text{ above } b\}, \quad \omega \mapsto \operatorname{cent}(E, \omega).$$

*Proof.* The group  $G$  acts transitively on the extensions of  $v_b$  to  $N$ , and the stabilizer of  $w_E$  is  $D_E$ . Restricting such a valuation to  $L = N^H$  identifies precisely those extensions whose indices differ on the right by an element of  $H$ . The resulting orbits are therefore indexed by  $D_E \backslash G/H$ , giving the stated bijection.

It remains to check independence of representatives. If  $D_E g' H = D_E g H$ , then  $g' = d g h$  for some  $d \in D_E$  and  $h \in H$ . The element  $d$  fixes  $w_E$ , while  $h$  fixes  $L$  pointwise, so

$$(g'^{-1} w_E)|_L = (g^{-1} w_E)|_L.$$

Moreover,  $I_E \trianglelefteq D_E$ , and conjugation by  $d$  identifies

$$I_E \cap g' H g'^{-1} \quad \text{with} \quad I_E \cap g H g^{-1}.$$

Thus  $u_\omega$ ,  $\operatorname{cent}(E, \omega)$ , and  $i_E(\omega)$  are well-defined. □

We use the notation

$$e(\operatorname{cent}(E, \omega)/b) := e(u_\omega/v_b).$$

Since the residue characteristic is zero,

$$\operatorname{Ram}^{(1)}(Z/Y) := \{E \in Z^{(1)} : e(w_E/v_b) > 1\} = \{E \in Z^{(1)} : I_E \neq 1\}.$$

**Proposition 1.10** (Detection of inertia on a conjugate sheet). *If  $I_E \neq 1$ , then there is an  $\omega \in \mathcal{Q}_E$  with*

$$i_E(\omega) > 1.$$



*Proof.* Suppose, to the contrary, that  $i_E(\omega) = 1$  for every  $\omega \in \mathcal{Q}_E$ . For each  $g \in G$ ,

$$1 = i_E(D_E g H) = [I_E : I_E \cap g H g^{-1}],$$

so

$$I_E = I_E \cap g H g^{-1} \subseteq g H g^{-1} \quad \text{for every } g \in G.$$

Consequently,

$$I_E \subseteq \bigcap_{g \in G} g H g^{-1} = \text{core}_G(H) = 1,$$

contrary to  $I_E \neq 1$ . □

**Proposition 1.11** (Intermediate ramification index). *For  $\omega = D_E g H \in \mathcal{Q}_E$ ,*

$$e(\text{cent}(E, \omega)/b) = i_E(\omega) = [I_E : I_E \cap g H g^{-1}].$$

*Proof.* Let  $w_E$  and  $v_b$  be the divisorial valuations associated with  $E$  and  $b$ , and let

$$w_g = g^{-1} w_E, \quad u_\omega = w_g|_L.$$

The valuation  $u_\omega$  has center  $\text{cent}(E, \omega)$  on  $\overline{X}$ . Since the residue fields have characteristic zero,

$$e(w_g/v_b) = |I_E|.$$

The inertia group of  $w_g$  over  $u_\omega$  is

$$g^{-1} I_E g \cap H,$$

and hence

$$e(w_g/u_\omega) = |g^{-1} I_E g \cap H| = |I_E \cap g H g^{-1}|.$$

Multiplicativity of ramification indices in the field tower

$$K \subseteq L \subseteq N$$

therefore gives

$$e(u_\omega/v_b) = \frac{e(w_g/v_b)}{e(w_g/u_\omega)} = [I_E : I_E \cap g H g^{-1}],$$

which is the asserted formula. □

**Proposition 1.12** (Visibility-to-boundary). *Suppose that  $X \rightarrow Y$  is étale. For  $E \in Z^{(1)}$  and  $\omega \in \mathcal{Q}_E$ ,*

$$i_E(\omega) > 1 \implies \text{cent}(E, \omega) \in \partial X.$$

*Proof.* Suppose that  $\text{cent}(E, \omega) \in j(X)$ , and let  $x \in X$  be the corresponding point under the open immersion  $j$ . The local homomorphism

$$\mathcal{O}_{Y,b} \longrightarrow \mathcal{O}_{X,x}$$

is étale, so  $e(\text{cent}(E, \omega)/b) = 1$ . Proposition 1.11 now gives

$$i_E(\omega) = e(\text{cent}(E, \omega)/b) = 1.$$

Taking the contrapositive gives the result. □

**Definition 1.13** (Hidden-inertia locus).

$$\mathcal{H}(\mathcal{D}) := \left\{ E \in \text{Ram}^{(1)}(Z/Y) : \begin{array}{l} \exists \omega \in \mathcal{Q}_E, \ i_E(\omega) > 1, \\ \forall \omega \in \mathcal{Q}_E, \ i_E(\omega) > 1 \implies \text{cent}(E, \omega) \in \partial X \end{array} \right\}.$$

Thus  $E \in \mathcal{H}(\mathcal{D})$  precisely when nontrivial inertia is visible on a conjugate sheet but every visible center belongs to the deleted boundary.

## 1.6 Automorphism detection

**Theorem 1.14** (Automorphism criterion). *For an arbitrary polynomial self-map  $F: \mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^n$ , the following are equivalent:*

- (1)  $F$  is a polynomial automorphism;
- (2)  $F$  is injective on  $\mathbb{A}^n(\mathbb{C})$ ;
- (3)  $I_R(F) = I_{\Delta}$ ;
- (4)  $\text{Obs}(F) = 0$ .

*Proof.* Suppose first that  $F$  has polynomial inverse  $G = (G_1, \dots, G_n)$ . Then

$$x_i - y_i = G_i(F(x)) - G_i(F(y)) \in I_R$$

for every  $i$ , so  $I_{\Delta} \subseteq I_R$ . Proposition 1.1 gives equality. Thus (1) implies (3).

Suppose  $I_R = I_{\Delta}$ , and let  $\text{ev}_{(a,b)}: S \rightarrow \mathbb{C}$  be evaluation at  $(a, b)$ . If  $F(a) = F(b)$ , then

$$I_R \subseteq \ker(\text{ev}_{(a,b)}).$$

Therefore  $x_i - y_i \in I_{\Delta} = I_R$  gives  $a_i - b_i = 0$  for every  $i$ . Thus  $a = b$ , proving (3) $\Rightarrow$ (2). The Ax–Grothendieck theorem [Ax69] gives (2) $\Rightarrow$ (1). Finally, Proposition 1.2 gives (3) $\iff$ (4).  $\square$

**Corollary 1.15** (The Jacobian conjecture as obstruction vanishing). *For every  $n \geq 1$ , the  $n$ -dimensional Jacobian conjecture is equivalent to each of the statements*

$$\forall F: \mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^n, \quad \det JF \in \mathbb{C}^{\times} \implies \text{Obs}(F) = 0,$$

and

$$\forall F: \mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^n, \quad \det JF \in \mathbb{C}^{\times} \implies I_R(F) = I_{\Delta}.$$

*Proof.* Apply Theorem 1.14 to every Keller map. Its equivalent conditions (1), (3), and (4) give the two displayed formulations.  $\square$

## 1.7 Generic degree one

**Corollary 1.16** (Generic degree-one descent). *If*

$$L = K \quad \text{and} \quad A \text{ is flat over } B,$$

*then*

$$\text{Obs}(F) = 0.$$

*Proof.* When  $L = K$ , Theorem 1.6 identifies the base-changed diagonal with tensor multiplication:

$$\ker(1 \otimes \bar{\mu}_F) \cong \ker\left(K \otimes_K K \xrightarrow{\mu_K} K\right) = 0.$$

The fiber-product identification of Theorem 1.3 is

$$C_F \xrightarrow{\sim} A \otimes_B A.$$

Since  $A$  is flat over  $B$ , the tensor product  $A \otimes_B A$ , and therefore  $C_F$ , is flat over  $B$ . The localization map

$$\iota_C: C_F \hookrightarrow K \otimes_B C_F$$

is therefore injective, and

$$\iota_C(\ker \bar{\mu}_F) \subseteq \ker(1 \otimes \bar{\mu}_F) = 0.$$

Thus  $\ker \bar{\mu}_F = 0$ . Proposition 1.2 identifies

$$\ker(\bar{\mu}_F) = \text{Obs}(F) = I_\Delta/I_R,$$

so  $\text{Obs}(F) = 0$ . □

*Lean correspondence.* `CollisionIdeals/CollisionDiagonal.lean`, `CollisionIdeals/OffDiagonalScheme.lean`, `CollisionIdeals/Secant.lean`, `CollisionIdeals/GenericCollisionDiagonal.lean`, `CollisionIdeals/KellerCollisionModel.lean`, and `CollisionIdeals/JacobianConjecture.lean`. The Lean form of Theorem 1.6 takes surjectivity of  $\theta$  as a parameter; the proof above derives it from generic finiteness.

## 2 Planar vanishing

Let

$$F = (P, Q): \mathbb{A}_{\mathbb{C}}^2 \longrightarrow \mathbb{A}_{\mathbb{C}}^2$$

be a polynomial map satisfying  $\det JF = c \in \mathbb{C}^\times$ .

### 2.1 The planar secant projector

Let  $\Delta = V(I_\Delta) \subseteq \text{Spec } S$ . For  $a \in S$ , write  $a|_\Delta$  for its image under  $S \rightarrow S/I_\Delta$ , and apply this notation entrywise to matrices. Let

$$\mathbf{d} = \begin{pmatrix} x_1 - y_1 \\ x_2 - y_2 \end{pmatrix}, \quad \mathbf{r} = \begin{pmatrix} P(x) - P(y) \\ Q(x) - Q(y) \end{pmatrix}.$$

For  $H \in \mathbb{C}[x_1, x_2]$ , let

$$D_1 H = \frac{H(x_1, x_2) - H(y_1, x_2)}{x_1 - y_1}, \quad D_2 H = \frac{H(y_1, x_2) - H(y_1, y_2)}{x_2 - y_2}.$$

Both quotients belong to  $S$ . Let

$$M_F = \begin{pmatrix} D_1 P & D_2 P \\ D_1 Q & D_2 Q \end{pmatrix}.$$

Then

$$\mathbf{r} = M_F \mathbf{d}, \quad M_F|_\Delta = JF.$$

Let  $\delta_F = \det(M_F)$ .

**Proposition 2.1** (Planar secant annihilation). *The secant determinant satisfies*

$$\delta_F I_\Delta \subseteq I_R, \quad \delta_F|_\Delta = c.$$

*Proof.* Multiplying  $\mathbf{r} = M_F \mathbf{d}$  by  $\text{adj}(M_F)$  gives

$$\delta_F \mathbf{d} = \text{adj}(M_F) \mathbf{r}.$$

$$\mathbf{r} \in I_R^{\oplus 2}, \quad \text{adj}(M_F) \in M_2(S) \implies \text{adj}(M_F) \mathbf{r} \in I_R^{\oplus 2}.$$

Therefore

$$\delta_F \mathbf{d} \in I_R^{\oplus 2},$$

or equivalently,

$$\forall j \in \{1, 2\}, \quad \delta_F(x_j - y_j) \in I_R.$$

Hence  $\delta_F I_\Delta \subseteq I_R$ . On the diagonal,

$$(D_j F_i)|_\Delta = \frac{\partial F_i}{\partial x_j},$$

so  $M_F|_\Delta = JF$  and  $\delta_F|_\Delta = \det JF = c$ . □

Let  $\bar{\delta}_F$  be the class of  $\delta_F$  in  $C_F$ , and define

$$q_F := 1 - c^{-1} \bar{\delta}_F.$$

**Theorem 2.2** (Planar collision-sheet decomposition). *The collision scheme decomposes as*

$$R_F \cong \Delta_{X/Y}(X) \amalg R_F^\circ, \quad C_F \cong A \times C_F^\circ.$$

*The off-diagonal ideal satisfies*

$$I_{\text{off}} = I_R + (\delta_F) = I_R : I_\Delta^\infty.$$

*Proof.* By Proposition 1.2,

$$J := \ker(\bar{\mu}_F) = I_\Delta / I_R = \text{Obs}(F).$$

Proposition 2.1 gives

$$\bar{\delta}_F J = 0, \quad \bar{\delta}_F - c \in J.$$

Theorem 1.5, applied to these two relations and  $q_F = 1 - c^{-1} \bar{\delta}_F$ , yields

$$q_F^2 = q_F, \quad J = C_F q_F, \quad \text{Ann}_{C_F}(J) = C_F(1 - q_F), \quad J = 0 \iff q_F = 0.$$

Because  $1 - q_F = c^{-1} \bar{\delta}_F$ ,

$$\frac{I_R : I_\Delta}{I_R} = \text{Ann}_{C_F}(J) = C_F(1 - q_F) = C_F \bar{\delta}_F = \frac{I_R + (\delta_F)}{I_R}.$$

Hence

$$I_{\text{off}} = I_R : I_\Delta = I_R + (\delta_F).$$

Idempotence gives  $J^m = C_F q_F = J$  for every  $m \geq 1$ . Therefore

$$\frac{I_R : I_\Delta^m}{I_R} = \text{Ann}_{C_F}(J^m) = \text{Ann}_{C_F}(J) = \frac{I_R : I_\Delta}{I_R}.$$

Thus

$$I_R : I_\Delta^\infty = \bigcup_{m \geq 1} (I_R : I_\Delta^m) = I_R : I_\Delta.$$

The idempotent  $q_F$  also gives a ring isomorphism

$$C_F \xrightarrow{\sim} C_F(1 - q_F) \times C_F q_F, \quad a \mapsto (a(1 - q_F), a q_F),$$

whose inverse is addition. The two factors satisfy

$$C_F(1 - q_F) \cong C_F / C_F q_F = C_F / J \xrightarrow[\bar{\mu}_F]{\sim} A$$

and

$$C_F q_F \cong C_F / C_F(1 - q_F) \cong S / (I_R + (\delta_F)) = C_F^\circ.$$

Thus  $C_F \cong A \times C_F^\circ$ . Since the spectrum of a product is the disjoint union of the two spectra, this isomorphism dualizes to

$$R_F \cong \Delta_{X/Y}(X) \amalg R_F^\circ.$$

□

Corollary 1.4 states that

$$R_F^\circ = \emptyset \iff C_F^\circ = 0 \iff \text{Obs}(F) = 0 \iff I_R = I_\Delta.$$

The preceding theorem identifies

$$C_F^\circ = S / (I_R + (\delta_F)),$$

and Theorem 1.5 gives  $\text{Obs}(F) = 0 \iff q_F = 0$ . Hence

$$q_F = 0 \iff I_R + (\delta_F) = S \iff R_F^\circ = \emptyset \iff I_R = I_\Delta.$$

In particular, the secant determinant cuts out the off-diagonal factor:

$$R_F^\circ = \text{Spec}(S / (I_R + (\delta_F))).$$

## 2.2 Boundary exclusion and inertia

Let

$$A = \mathbb{C}[x_1, x_2], \quad B = \mathbb{C}[P, Q] \subseteq A, \quad K = \mathbb{C}(P, Q), \quad L = \mathbb{C}(x_1, x_2),$$

and let  $X = \text{Spec } A$  and  $Y = \text{Spec } B$ .

**Proposition 2.3** (Keller local geometry).

$$\mathbb{C}[u, v] \xrightarrow{\sim} B, \quad u \mapsto P, \quad v \mapsto Q.$$

Moreover,

$$X \xrightarrow{f_F} Y \text{ is étale,} \quad A \text{ is flat over } B, \quad [L : K] < \infty \text{ and } L/K \text{ is separable.}$$

*Proof.* The identity

$$dP \wedge dQ = \left( \frac{\partial P}{\partial x_1} \frac{\partial Q}{\partial x_2} - \frac{\partial P}{\partial x_2} \frac{\partial Q}{\partial x_1} \right) dx_1 \wedge dx_2 = \det(JF) dx_1 \wedge dx_2 = c dx_1 \wedge dx_2 \neq 0$$

shows that  $P, Q$  are algebraically independent. Thus  $B \cong \mathbb{C}[u, v]$  and  $Y \cong \mathbb{A}_{\mathbb{C}}^2$ . Moreover,

$$A \cong B[U, V]/(P(U, V) - u, Q(U, V) - v),$$

whose relative Jacobian determinant is  $c \in A^\times$ . The Jacobian criterion therefore makes  $B \rightarrow A$  étale. In particular it is flat and quasi-finite. Since  $f_F$  is dominant and quasi-finite, its generic fiber is zero-dimensional, so  $[L : K] < \infty$ . The morphism remains étale at the generic point; hence the finite extension  $L/K$  is separable.  $\square$

Fix an algebraic closure  $\overline{K} \supseteq L$ , and let

$$\begin{aligned} N &:= K(\sigma(L) : \sigma \in \text{Hom}_K(L, \overline{K})), & K &\subseteq L \subseteq N \subseteq \overline{K}, \\ \overline{A} &:= \{a \in L : a \text{ is integral over } B\}, & A_N &:= \{a \in N : a \text{ is integral over } B\}, \\ \overline{X} &:= \text{Spec } \overline{A}, & Z &:= \text{Spec } A_N. \end{aligned}$$

Thus  $N/K$  is the normal closure of  $L/K$ .

**Proposition 2.4** (Planar finite-cover diagram).

$$Z \xrightarrow[\text{finite}]{\pi} \overline{X} \xrightarrow[\text{finite}]{\bar{\nu}} Y, \quad \nu = \bar{\nu} \circ \pi,$$

and

$$X \xrightarrow[\text{open}]{j} \overline{X}, \quad \bar{\nu} \circ j = f_F.$$

*Proof.* Since  $B \cong \mathbb{C}[u, v]$  is a finitely generated normal  $\mathbb{C}$ -algebra, its integral closures in the finite extensions  $L/K$  and  $N/K$  are finite over  $B$ . The inclusion  $L \subseteq N$  gives  $\overline{A} \subseteq A_N$ , so

$$B \subseteq \overline{A} \subseteq A_N.$$

The morphism  $\bar{\nu}$  is finite because  $\overline{A}$  is a finite  $B$ -module. If  $a_1, \dots, a_r$  generate  $A_N$  over  $B$ , then they also generate  $A_N$  over  $\overline{A}$ , since  $B \subseteq \overline{A}$ . Hence  $\pi$  is finite.

For  $a \in \overline{A}$ , integrality over  $B \subseteq A$  and the normality of  $A = \mathbb{C}[x_1, x_2]$  give  $a \in A$ . Thus

$$B \subseteq \overline{A} \subseteq A, \quad \text{Frac}(\overline{A}) = \text{Frac}(A) = L,$$

induces a dominant birational morphism  $j : X \rightarrow \overline{X}$ . It is quasi-finite and separated by Proposition 2.3. Since  $\overline{X}$  is normal, Zariski's Main Theorem makes  $j$  an open immersion.  $\square$

Let

$$\begin{aligned} \mathcal{D}_F &= \left( Z \xrightarrow{\pi} \overline{X} \xrightarrow{\bar{\nu}} Y, j : X \hookrightarrow \overline{X} \right), & \partial X &= \overline{X} \setminus j(X), \\ G &= \text{Gal}(N/K), & H &= \text{Gal}(N/L). \end{aligned}$$

For  $E \in \text{Ram}^{(1)}(Z/Y)$  and  $\omega \in \mathcal{Q}_E$ , Proposition 1.11 gives

$$e(\text{cent}(E, \omega)/\nu(E)) = i_E(\omega).$$

Since  $X \rightarrow Y$  is étale by Proposition 2.3, Proposition 1.12 gives

$$i_E(\omega) > 1 \implies \text{cent}(E, \omega) \in \partial X.$$

Thus Definition 1.13 specializes to

$$\mathcal{H}(\mathcal{D}_F) = \left\{ E \in \text{Ram}^{(1)}(Z/Y) : \begin{array}{l} \exists \omega \in \mathcal{Q}_E, \ i_E(\omega) > 1, \\ \forall \omega \in \mathcal{Q}_E, \ i_E(\omega) > 1 \implies \text{cent}(E, \omega) \in \partial X \end{array} \right\}.$$

## 2.3 Hidden inertia and divisorial ramification

**Corollary 2.5** (Planar divisorial ramification criterion).

$$\mathcal{H}(\mathcal{D}_F) = \emptyset \iff \text{Ram}^{(1)}(Z/Y) = \emptyset.$$

*Proof.* The reverse implication follows from  $\mathcal{H}(\mathcal{D}_F) \subseteq \text{Ram}^{(1)}(Z/Y)$ . Conversely, suppose that  $\mathcal{H}(\mathcal{D}_F) = \emptyset$  and let  $E \in \text{Ram}^{(1)}(Z/Y)$ . Then  $I_E \neq 1$ . Since  $N$  is the normal closure of  $L/K$ ,  $\text{core}_G(H) = 1$ , and Proposition 1.10 gives

$$\exists \omega \in \mathcal{Q}_E, \quad i_E(\omega) > 1.$$

For every  $\omega' \in \mathcal{Q}_E$ , Proposition 1.12 gives

$$i_E(\omega') > 1 \implies \text{cent}(E, \omega') \in \partial X.$$

Hence  $E \in \mathcal{H}(\mathcal{D}_F)$ , a contradiction.  $\square$

## 2.4 Collapse of the normal closure

**Theorem 2.6** (Purity of the branch locus, specialized). *Let  $Z \rightarrow \mathbb{A}_{\mathbb{C}}^2$  be a finite dominant generically separable morphism with  $Z$  integral and normal. If  $\text{Ram}^{(1)}(Z/\mathbb{A}_{\mathbb{C}}^2) = \emptyset$ , then the morphism is étale [The26].*

**Theorem 2.7** (Finite-étale rigidity of the affine plane). *Every connected finite étale cover of  $\mathbb{A}_{\mathbb{C}}^2$  is trivial [GR71, Exposé XII].*

**Corollary 2.8** (Triviality of the planar normal closure). *If*

$$\text{Ram}^{(1)}(Z/Y) = \emptyset,$$

*then*

$$N = K \quad \text{and hence} \quad L = K.$$

*Proof.* By Proposition 2.3,  $Y \cong \mathbb{A}_{\mathbb{C}}^2$ . Since  $N/K$  is finite Galois,  $\nu: Z \rightarrow Y$  is generically separable; hence Theorem 2.6 makes it finite étale. The inclusion

$$A_N \subset N \text{ is a domain,} \quad \text{so} \quad Z = \text{Spec } A_N \text{ is connected.}$$

Theorem 2.7 therefore makes  $\nu: Z \rightarrow Y$  an isomorphism. Taking function fields gives  $N = K$ . Finally,  $K \subseteq L \subseteq N$  implies  $L = K$ .  $\square$

## 2.5 Vanishing of the off-diagonal factor

**Theorem 2.9** (Planar Vanishing). *For every polynomial map  $F = (P, Q): \mathbb{A}_{\mathbb{C}}^2 \rightarrow \mathbb{A}_{\mathbb{C}}^2$  satisfying  $\det JF = c \in \mathbb{C}^\times$ ,*

$$\mathcal{H}(\mathcal{D}_F) = \emptyset \implies \begin{array}{ll} I_R = I_\Delta, & \text{Obs}(F) = 0, \\ q_F = 0, & R_F^\circ = \emptyset. \end{array}$$

*Proof.* Corollary 2.5 gives  $\text{Ram}^{(1)}(Z/Y) = \emptyset$ , and Corollary 2.8 gives  $L = K$ . Proposition 2.3 gives that  $A$  is flat over  $B$ . Corollary 1.16 states

$$L = K, \quad A \text{ flat over } B \implies \text{Obs}(F) = 0,$$

so  $\text{Obs}(F) = 0$ .

Corollary 1.4 and Theorem 1.5 state

$$\begin{aligned} R_F^\circ = \emptyset &\iff C_F^\circ = 0 \iff \text{Obs}(F) = 0 \iff I_R = I_\Delta, \\ \text{Obs}(F) = 0 &\iff q_F = 0. \end{aligned}$$

All four conclusions follow.  $\square$

**Corollary 2.10** (Conditional Jacobian conjecture in dimension two). *If  $\mathcal{H}(\mathcal{D}_F) = \emptyset$  for every planar Keller map  $F$ , then every planar Keller map is a polynomial automorphism. Equivalently, the two-dimensional Jacobian conjecture holds.*

*Proof.* The automorphism criterion, Theorem 1.14, states that every polynomial self-map of  $\mathbb{A}_{\mathbb{C}}^2$  satisfies

$$I_R = I_\Delta \iff F \text{ is a polynomial automorphism.}$$

Planar Vanishing supplies the left-hand side for every planar Keller map under the stated condition.  $\square$

**External inputs to the formalization.** The Lean development takes the following established results as explicit inputs:

- (E1) The planar Ax–Grothendieck theorem: every injective polynomial self-map of  $\mathbb{A}_{\mathbb{C}}^2$  is a polynomial automorphism [Ax69].
  - (E2) Purity of the branch locus in the specialized form of Theorem 2.6 [The26].
  - (E3) Every connected finite étale cover of  $\mathbb{A}_{\mathbb{C}}^2$  is trivial, as in Theorem 2.7 [GR71, Exposé XII].
- The condition  $\mathcal{H}(\mathcal{D}_F) = \emptyset$  is not included among these inputs; it remains the explicit premise of Theorem 2.9.

*Lean correspondence.* The conditional planar chain is collected in `CollisionIdeals/Planar.lean`. The three inputs above are isolated in `CollisionIdeals/Planar/ExternalAssumptions.lean`.

### 3 The cubic $S_3$ obstruction in dimension three

Let  $S_3 = \text{Perm}(\{1, 2, 3\})$  be the symmetric group on three letters.

**Proposition 3.1** (Dimension-three counterexample criterion). *For a polynomial map*

$$F: \mathbb{A}_{\mathbb{C}}^3 \longrightarrow \mathbb{A}_{\mathbb{C}}^3,$$

*consider the following statements:*

- (1)  $\det JF \in \mathbb{C}^\times$ ;
- (2)  $F$  is not a polynomial automorphism;



(3)  $R_F^\circ \neq \emptyset$ .

Under (1), statements (2) and (3) are equivalent.

*Proof.* By Corollary 1.4 and Theorem 1.14,

$$R_F^\circ = \emptyset \iff I_R = I_\Delta \iff F \in \text{Aut}_{\text{poly}}(\mathbb{A}_{\mathbb{C}}^3).$$

Negating this equivalence proves (2)  $\iff$  (3). □

### 3.1 The cubic extension and its normal closure

Let

$$F = (F_1, F_2, F_3): \mathbb{A}_{\mathbb{C}}^3 \longrightarrow \mathbb{A}_{\mathbb{C}}^3$$

be dominant and generically finite. Let

$$A = \mathbb{C}[x_1, x_2, x_3], \quad B = \mathbb{C}[F_1, F_2, F_3] \subseteq A, \quad K = \text{Frac}(B), \quad L = \text{Frac}(A),$$

and let

$$\theta: K \otimes_B A \longrightarrow L, \quad \lambda \otimes a \longmapsto \lambda a,$$

be the isomorphism of Theorem 1.6. Suppose that  $L/K$  is separable and nonnormal, and choose a power basis with generator  $\alpha$  and degree three:

$$L = K(\alpha), \quad [L : K] = 3.$$

If  $m_\alpha \in K[T]$  is the minimal polynomial of  $\alpha$ , let  $h_\alpha \in L[T]$  be determined by

$$m_\alpha(T) = (T - \alpha)h_\alpha(T) \quad \text{in } L[T].$$

Let  $N/K$  be the normal closure of  $L/K$ , containing  $L$  through the marked inclusion. Let

$$G = \text{Gal}(N/K), \quad H = \text{Gal}(N/L).$$

Recall from Section 1.5 that

$$G/H \xrightarrow{\sim} \text{Hom}_K(L, N), \quad gH \longmapsto g|_L, \quad \text{core}_G(H) = 1.$$

The final equality follows because  $N$  is the normal closure of  $L/K$ ; equivalently, the action of  $G$  on the conjugate embeddings of  $L$  is faithful.

For  $Y = \text{Spec } B$ , the finite-normalization construction of Section 1.5 gives

$$Z = \text{Norm}_N(Y) \longrightarrow \overline{X} = \text{Norm}_L(Y) \longrightarrow Y.$$

If  $E \in Z^{(1)}$ , with decomposition group  $D_E$  and inertia group  $I_E$ , then its conjugate centers are indexed by  $\mathcal{Q}_E = D_E \backslash G/H$ . For  $\omega = D_E gH$ , Proposition 1.11 states

$$e(\text{cent}(E, \omega)/\nu(E)) = i_E(\omega) = [I_E : I_E \cap gHg^{-1}].$$

The following proposition is the standard residual-factor description of a separable nonnormal cubic extension. We include its short proof to fix the marked identification with  $N$ .

**Proposition 3.2** (Cubic residual-factor identification). *The polynomial  $h_\alpha$  is irreducible over  $L$ , and there is an  $L$ -algebra isomorphism*

$$\varphi: L[T]/(h_\alpha) \xrightarrow{\sim}_L N.$$

Consequently,

$$\Psi: L \otimes_K L \xrightarrow{\sim}_L L \times N, \quad \text{pr}_1 \circ \Psi = \mu_L.$$

*Proof.* Write

$$m_\alpha(T) = T^3 + a_2T^2 + a_1T + a_0.$$

If the quadratic  $h_\alpha$  were reducible over  $L$ , it would have a root  $\beta \in L$ . Separability gives  $\beta \neq \alpha$ , and the third root would be

$$\gamma = -a_2 - \alpha - \beta \in L.$$

Thus  $m_\alpha$  would split over  $L$ . Since  $L = K(\alpha)$ , the extension  $L/K$  would then be the splitting field of the separable polynomial  $m_\alpha$ , hence normal. This contradicts the assumed nonnormality, so  $h_\alpha$  is irreducible.

Let  $\beta$  denote the image of  $T$  in  $L[T]/(h_\alpha)$ . This field contains the three roots

$$\alpha, \quad \beta, \quad -a_2 - \alpha - \beta$$

of  $m_\alpha$ , and is generated over  $L$  by  $\beta$ . It is therefore the splitting field of  $m_\alpha$ , which is the normal closure  $N$  of  $L/K$ . Choose the resulting  $L$ -algebra isomorphism

$$\varphi: L[T]/(h_\alpha) \xrightarrow{\sim}_L N.$$

Let

$$\Phi_\alpha: L \otimes_K L \xrightarrow{\sim}_L L \times L[T]/(h_\alpha)$$

be the marked-root decomposition of Theorem 1.7. It satisfies

$$\text{pr}_1 \circ \Phi_\alpha = \mu_L.$$

Composing its second factor with  $\varphi$ , let

$$\Psi = (\text{id}_L \times \varphi) \circ \Phi_\alpha.$$

Since  $\text{id}_L \times \varphi$  leaves the first factor unchanged,  $\text{pr}_1 \circ \Psi = \mu_L$ . □

**Theorem 3.3** (Galois group of a nonnormal cubic extension).

$$\text{Gal}(N/K) \cong S_3.$$

*Proof.* Proposition 3.2 identifies  $N$  with the degree-two extension  $L[T]/(h_\alpha)$  of  $L$ . Hence

$$|H| = [N : L] = 2.$$

The normal-closure identities recalled above give

$$\text{core}_G(H) = 1, \quad [G : H] = [L : K] = 3.$$

Hence the action of  $G$  on the three cosets of  $H$  gives

$$\rho: G \longrightarrow \text{Perm}(G/H) \cong S_3, \quad \ker \rho = \text{core}_G(H) = 1.$$

Core-freeness therefore makes  $\rho$  injective, so

$$|G| \leq |S_3| = 6.$$

The index formula gives

$$|G| = [G : H]|H| = 3 \cdot 2 = 6.$$

Thus  $|G| = 6$ , and the injective homomorphism  $\rho$  is an isomorphism  $G \cong S_3$ .  $\square$

### 3.2 The cubic off-diagonal factor

Retain the cubic data of Section 3.1, and suppose in addition that  $F$  is Keller. Thus

$$B = \mathbb{C}[F_1, F_2, F_3] \subseteq A = \mathbb{C}[x_1, x_2, x_3], \quad K = \text{Frac}(B) \subseteq L = \text{Frac}(A) = K(\alpha) \subseteq N,$$

where  $L/K$  is separable, nonnormal, and of degree three, and  $N/K$  is its normal closure. Let

$$\theta: K \otimes_B A \xrightarrow{\sim} L, \quad \lambda \otimes a \longmapsto \lambda a.$$

**Corollary 3.4** (Generic cubic  $S_3$  collision). *There is a  $K$ -algebra isomorphism*

$$\Psi_F: K \otimes_B C_F \xrightarrow{\sim}_K L \times N, \quad \text{pr}_1 \circ \Psi_F = \theta \circ (1 \otimes \bar{\mu}_F).$$

*It satisfies*

$$\Psi_F(\ker(\theta \circ (1 \otimes \bar{\mu}_F))) = 0 \times N, \quad \text{Gal}(N/K) \cong S_3.$$

*Consequently,*

$$\text{Obs}(F) \neq 0, \quad I_R \subsetneq I_\Delta, \quad R_F^\circ \neq \emptyset, \quad F \notin \text{Aut}_{\text{poly}}(\mathbb{A}_{\mathbb{C}}^3).$$

*Proof.* Theorem 1.6 gives a  $K$ -algebra isomorphism

$$\Phi_F: K \otimes_B C_F \xrightarrow{\sim}_K L \otimes_K L$$

such that

$$\mu_L \circ \Phi_F = \theta \circ (1 \otimes \bar{\mu}_F).$$

Proposition 3.2 gives

$$\Psi: L \otimes_K L \xrightarrow{\sim}_L L \times N, \quad \text{pr}_1 \circ \Psi = \mu_L.$$

After restricting  $\Psi$  from  $L$  to  $K$ , let

$$\Psi_F = \Psi \circ \Phi_F.$$

It follows that

$$\text{pr}_1 \circ \Psi_F = \theta \circ (1 \otimes \bar{\mu}_F), \quad \Psi_F(\ker(\theta \circ (1 \otimes \bar{\mu}_F))) = 0 \times N \neq 0.$$

Suppose that the diagonal-evaluation homomorphism  $\bar{\mu}_F: C_F \rightarrow A$  were injective. Since  $K$  is a localization of  $B$ , it is flat over  $B$ , so  $1 \otimes \bar{\mu}_F$  would also be injective. Theorem 1.6 shows that  $\theta$  is an isomorphism. Therefore  $\theta \circ (1 \otimes \bar{\mu}_F)$  would be injective, contradicting the nonzero kernel  $0 \times N$  above. Hence  $\ker(\bar{\mu}_F) \neq 0$ , and Proposition 1.2 gives  $\text{Obs}(F) \neq 0$ . Proposition 1.1 shows that  $I_R \subseteq I_\Delta$  strict, and Corollary 1.4 gives  $R_F^\circ \neq \emptyset$ .

Finally, Theorem 3.3 identifies  $\text{Gal}(N/K)$  with  $S_3$ . If  $F$  were a polynomial automorphism, Theorem 1.14 would give  $\text{Obs}(F) = 0$ , contrary to  $\text{Obs}(F) \neq 0$ . Since  $F$  is Keller, it is a dimension-three Jacobian-conjecture counterexample.  $\square$

For the same cubic map and field tower, let

$$X = \operatorname{Spec} A, \quad Y = \operatorname{Spec} B, \quad G = \operatorname{Gal}(N/K), \quad H = \operatorname{Gal}(N/L).$$

Suppose there is a diagram over  $Y$

$$\mathcal{D} = \left( Z = \operatorname{Norm}_N(Y) \xrightarrow{\pi} \overline{X} = \operatorname{Norm}_L(Y) \xrightarrow{\bar{\nu}} Y, j: X \hookrightarrow \overline{X} \right)$$

in which  $\pi$  and  $\bar{\nu}$  are finite,  $j$  is an open immersion,  $\bar{\nu} \circ j = f_F$ , and  $f_F: X \rightarrow Y$  is étale. Let

$$\partial X = \overline{X} \setminus j(X), \quad G \cong S_3$$

by Theorem 3.3.

**Corollary 3.5** (Hidden inertia in the  $S_3$ -normalization).

$$\operatorname{Ram}^{(1)}(Z/Y) \subseteq \mathcal{H}(\mathcal{D}).$$

*Proof.* Fix

$$E \in \operatorname{Ram}^{(1)}(Z/Y).$$

By the characteristic-zero description of divisorial ramification,  $I_E \neq 1$ . Because  $N$  is the normal closure of  $L/K$ ,  $\operatorname{core}_G(H) = 1$ . Proposition 1.10 states in this case that

$$\exists \omega \in \mathcal{Q}_E, \quad i_E(\omega) > 1.$$

Moreover, Proposition 1.12, applied to the étale map  $f_F$ , states that for every  $\omega' \in \mathcal{Q}_E$ ,

$$i_E(\omega') > 1 \implies \operatorname{cent}(E, \omega') \in \partial X.$$

These are exactly the existence and boundary conditions in Definition 1.13; hence  $E \in \mathcal{H}(\mathcal{D})$ . Since  $E$  was arbitrary, the stated inclusion follows.  $\square$

*Remark* (Ordered roots). Let

$$m_\alpha(T) = T^3 + a_2T^2 + a_1T + a_0$$

have roots  $\alpha_1, \alpha_2, \alpha_3$  in  $N$ . For distinct  $i, j, k$ ,

$$\alpha_k = -a_2 - \alpha_i - \alpha_j, \quad K(\alpha_i, \alpha_j) = K(\alpha_1, \alpha_2, \alpha_3) = N.$$

Thus the off-diagonal factor, which marks two distinct roots, is already the ordered-root normal closure.

*Lean correspondence.* `CollisionIdeals/ComplexThree/CubicGaloisGroup.lean`, `CollisionIdeals/ComplexThree/S3Collision.lean`, `CollisionIdeals/ComplexThree/OffDiagonal.lean`, `CollisionIdeals/ComplexThree/CubicS3.lean`, and `CollisionIdeals/ComplexThree/JacobianConjecture.lean`. The Lean cubic witness takes the residual-field isomorphism and  $H \neq 1$  as explicit inputs; Proposition 3.2 derives them here from nonnormality. Hidden inertia is formalized pointwise at a supplied ramified divisor.

## 4 Conclusion

The quotient  $\text{Obs}(F)$  vanishes exactly when equality of images agrees scheme-theoretically with equality of points. In dimension two,  $\mathcal{H}(\mathcal{D}_F) = \emptyset$  implies  $\text{Obs}(F) = 0$ ; in the cubic dimension-three class, its nonzero generic factor is the ordered-root  $S_3$ -normal closure. Both statements arise from the same collision, off-diagonal, and conjugate-sheet constructions.

Under the respective conditions of Sections 2 and 3, and with  $\text{Ram}^{(1)}(Z/Y) \neq \emptyset$  in the cubic normalization, the contrast is

| dimension two                    | dimension three                             |
|----------------------------------|---------------------------------------------|
| hidden inertia excluded          | hidden inertia in the $S_3$ -cover realized |
| $\text{Obs}(F) = 0$              | $\text{Obs}(F) \neq 0$                      |
| $I_R = I_\Delta$                 | $I_R \subsetneq I_\Delta$                   |
| $F$ is a polynomial automorphism | $F$ is not a polynomial automorphism.       |

**Motivating example.** The explicit three-dimensional polynomial counterexample announced by Levent Alpöge and independently formalized in Isabelle/HOL by Ramos, Hulak, and de Queiroz [RHdQ26] motivated the cubic  $S_3$  comparison. The formal verification establishes a constant nonzero Jacobian determinant and distinct rational points with the same image. That particular map is not used in the arguments above: the dimension-three result is the structural implication from the stated cubic and marked-normal-closure conditions.

The formalization uses Lean 4 and Mathlib [dMU21, The20].

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